

Unit 3, Lesson 1: Product Laws of Exponents



Benchmark

MA.7.NSO.1.1 Know and apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational number bases.



Learning Target

- ☐ I can explain and use the power of a product, power of a power, product of powers, and identity laws of exponents.



3.1.1 Warm-Up: Mistakes Help You Grow

- Two artists are drawing a plant. As they go through the process, both artists make mistakes and start over on their drawings. One of the artists crumples their papers up and throws them on the floor. The other artist hangs them on the wall next to where she is working.



What do you think is the purpose of hanging up the drawings with mistakes?



3.1.2 Exploration: Product of Powers Exponent Law

Recall that repeated multiplication is a way of writing a number with an exponent.

The number 2, called the **base**, is being multiplied repeatedly.

$$2^3 = 2 \times 2 \times 2$$

base $\rightarrow$ 2 is being multiplied repeatedly



A **base** is the number used as a factor in exponential form.

Since the base, 2, is a factor 3 times, the **exponent** is 3.

$$2^3 = 2 \times 2 \times 2$$

exponent $\rightarrow$ 2 is a factor 3 times



An **exponent** is the number of times the base occurs as a factor.

1. With your partner, use the definition of an exponent to complete the table.
 - a. Write each product as a repeated multiplication expression. Then, write each expression with an exponent.

Product	Repeated Multiplication	Power of the Form a^b
$2^2 \cdot 2^3$	$(2 \cdot 2) \cdot (2 \cdot 2 \cdot 2) = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$	2^5
$3^5 \cdot 3^4$		
$5^1 \cdot 5^4 \cdot 5^3$		

- b. How do the exponents in the first column relate to the exponents in the last column?
 - c. What do you notice about the bases in all three columns?
2. Write a rule for the product of two exponential expressions with the same base.



Product of Powers Exponent Law

When multiplying two powers that have the same nonzero base, _____ the exponents and keep the same base.

3. With your partner, follow the steps.

- Determine which column you will start in, and which column your partner will start in.
- Using the directions for your column, evaluate the expression. Your partner should use the directions in their column to evaluate the same expression.
- Verify if your answers are the same. If not, find the error(s).
- After each problem, switch columns.

The first expression has been evaluated.

Evaluate each power, and then multiply.	Apply the product of powers exponent law, and then evaluate the power.
$7^2 \cdot 7^1$	
$7^2 \cdot 7^1$	$7^2 \cdot 7^1$
$49 \cdot 7$	$7^{(2+1)}$
343	7^3
	343
$6^9 \cdot 6^3$	
$(-8)^4 \cdot (-8)^5$	

4. Julian says that he can evaluate $3^2 \cdot 2^3$ using the product of powers exponent law.

a. Explain to Julian if he is correct.

b. Complete the statement about the product of powers exponent law.

When a is a nonzero real number, and m and n are integers, the exponential expression $a^m \cdot a^n$ is equal

to

☐ a^{mn} .
☐ a^{m-n} .
☐ a^{m+n} .



3.1.3 Exploration: Power of a Product Exponent Law

- With your partner, complete the table and answer the questions.
 - Write each power of a product as a repeated multiplication expression, and group repeated factors by using the commutative and associative properties. Then, write each expression with an exponent in the form $a^m \cdot b^n$.

Power of a Product	Repeated Multiplication	Group Repeated Factors	Power of the Form $a^m \cdot b^n$
$(2 \cdot 3)^3$	$(2 \cdot 3) \cdot (2 \cdot 3) \cdot (2 \cdot 3)$	$(2 \cdot 2 \cdot 2) \cdot (3 \cdot 3 \cdot 3)$	$2^3 \cdot 3^3$
$(6 \cdot 5)^4$			
$(10 \cdot 4)^2$			

- How does the base in the first column relate to the bases in the last column?
 - How does the exponent in the first column relate to the exponents in the last column?
- Write a rule for the power of a product.



Power of a Product Exponent Law

When raising a product to a power, distribute the power to each factor.

- With your partner, follow the steps.
 - Determine which column you will start in, and which column your partner will start in.
 - Using the directions for your column, evaluate the expression. Your partner should use the directions in their column to evaluate the same expression.
 - Verify if your answers are the same. If not, find the error(s).
 - After each problem, switch columns.

The first expression has been evaluated.

Evaluate inside the parentheses, then apply the exponent.	Apply the power of a product exponent law, then evaluate the power.
$(^{-}5 \cdot 6)^2$	
$(^{-}5 \cdot 6)^2$	$(^{-}5 \cdot 6)^2$
$(^{-}30)^2$	$(^{-}5)^2 \cdot 6^2$
$(^{-}30) \cdot (^{-}30)$	$25 \cdot 36$
900	900
$(^{-}6 \cdot 4)^5$	
$(^{-}3 \cdot ^{-}2)^8$	

4. Aiyanna says that she can evaluate $(10 + -2)^3$ using the power of a product exponent law. Explain to Aiyanna if she is correct.

5. Complete the statement about the power of a product exponent law.

When a and b are nonzero real numbers, and m is an integer, the exponential expression $(a \cdot b)^m$ is equal

to

- a^{bm} .
- $a^m \cdot b^m$.
- $a^m + b^m$.



3.1.4 Exploration: Power of a Power Exponent Law

1. With your partner, complete the table.
 - a. Write each power of a power as a repeated multiplication expression using the outer exponent. Then, using the Product of Powers Exponent Law, rewrite the multiplication expression in the form a^m .

Power of a Power	Repeated Multiplication	Power of the Form a^m
$(4^2)^3$	$4^2 \cdot 4^2 \cdot 4^2$	4^6
$(10^4)^4$		
$(3^1)^5$		

- b. How do the exponents in the first column relate to the exponents in the last column?

2. Two different methods for evaluating expressions with exponents are illustrated in the table. With your partner, follow the steps.
 - Select which column you will start in, and which column your partner will start in.
 - Review Method 1 and Method 2. When you are working in each column, make sure to follow the same process as in the example for that column.
 - After finding the value of each expression, verify your answers are the same. If not, find the error(s).
 - After each problem, swap columns so that you and your partner are practicing both methods.

Example	Method 1	Method 2
a.	$[(-2)^4]^2$	
	$[(-2)^4]^2$	$[(-2)^4]^2$
	$(16)^2$	$(-2)^{4 \cdot 2}$
	$16 \cdot 16$	$(-2)^8$
	256	256
b.	$[(-6)^1]^8$	
c.	$[(-10)^2]^6$	

3. Complete the statement.

When a is a nonzero real number, and m and n are integers, the exponential expression $(a^m)^n$ is equal

to

- a^{m+n} .
- $a^m \cdot a^m$.
- $a^{m \times n}$.



Power of a Power Exponent Law

When raising a base with a power to another power, multiply the exponents and keep the same base.



3.1.5 Exploration: Identity Exponent Law

- With your partner, use the meaning of an exponent to complete the table. Write each product as a repeated multiplication expression, then write the value of each expression.

Exponential Expression	Repeated Multiplication	Value
10^1	10	10
100^1		
11^1		

- Complete the statement about the identity exponent law.

When a is a real number, the exponential expression a^1

is equal to

- 1.
- $a + 1$.
- a .



Identity Exponent Law

When raising a base to the power of 1, the base is a factor one time, thus, has the same value as the base.



3.1.6 Your Turn!

- Complete the following steps for each expression.
 - Rewrite each expression shown by using the laws of exponents.
 - Choose the exponent law you used from the list.
 - Evaluate the exponential expression.

$(2^3)^4$	☐ product of powers law ☐ power of product law ☐ power of power law ☐ identity exponent law
$(2^3)^4 = \underline{\hspace{2cm}}$	

$0.5^4 \cdot 0.5^3$	☐ product of powers law ☐ power of product law ☐ power of power law ☐ identity exponent law
$0.5^4 \cdot 0.5^3 = \underline{\hspace{2cm}}$	

$\left(\frac{24}{8}\right)^1$	☐ product of powers law ☐ power of product law ☐ power of power law ☐ identity exponent law
$\left(\frac{24}{8}\right)^1 = \underline{\hspace{2cm}}$	

$(0.5 \cdot ^{-}8)^2$	☐ product of powers law ☐ power of product law ☐ power of power law ☐ identity exponent law
$(0.5 \cdot ^{-}8)^2 = \underline{\hspace{2cm}}$	



3.1.7 Wrap-Up: Ticket Out the Door

- Rewrite the expressions using laws of exponents. State the exponent law used.

Expression	Exponent Law	Equivalent Expression
$12.75^5 \cdot 12.75^4$		
$\left[\left(\frac{1}{4}\right)^7\right]^2$		
$(12 \cdot 0.75)^3$		
285.6		
$\left(\frac{-3}{5}\right)^1$		
$10^6 \cdot (-1.5)^6$		